

Product overview

Potential relay is designed to start single-phase motors which employ both start and run capacitor(CSR and CSIR configurations) for high starting torque.relays are manufactured for the HVAC industry. These relays meet all requirements for original equipment relays.

Features&benefits

- ◇50A switching capability
- ◇Has a variety of mounting positions
- ◇With a variety of wiring
- ◇Dust cover type packaging structure
- ◇Used on heat pumps,vendingmachines,fans,and heating and heating and air conditioning

Product certification

c UL us	UL	UL 60947-4-1
RoHS	RoHS	RoHS Compliant

Code explanation

JPR	550/	AA	1	(XXX)

Type

Coil number
549/550/554/555/558/556/559/560/561

Operation characteristics
AA to AS (See table for operating characteristics)

Operation position
1/2/3/4/5/6

Special code1
XXX: Customer special requirement

Product model and specification

OPERATING CHARACTERISTICS AT 60Hz

H.P.U. = Approximate Pick-up at 90°C P.U. and D.O. values at 25°C

Coil number	549		550		554		555		558		556		559		560		561	
Vmax at 40°C (VAC)	170		332		336		375		379		395		420		450		500	
Resistance at 25°C (Ω) ±10%	1500		5600		5700		7500		7600		10000		10700		11000		13800	
	H.P.U.	P.U.	D.O.	P.U.	D.O.	P.U.	D.O.	P.U.	D.O.	P.U.	D.O.	P.U.	D.O.	P.U.	D.O.	P.U.	D.O.	P.U.	D.O.
AA	120-130	111-124	20-45												20-45				
AB	130-140	120-134	20-45	132-148	40-90														
AC	150-160	130-144	20-45	140-153	40-90														
AD	160-170	140-153	20-45	150-163	40-90														
AE	170-180	149-163	20-45	162-175	40-90														
AF	180-190			171-184	40-90	162-175	40-90					180-195	40-105						
AG	190-200			180-193	40-90	171-184	40-90	180-195	40-105			189-205	40-105	180-195	60-121				
AH	200-220			186-215	40-90	180-193	40-90	190-215	40-105	180-195	40-105	186-214	60-130	189-224	60-121				
AI	220-240			205-234	40-90	186-215	40-90	208-239	50-110	190-215	40-105	204-233	60-130	204-233	60-121				
AL	240-260			224-252	40-105	205-234	40-90	224-252	50-110	208-239	50-110	223-252	60-130	223-259	60-121	223-252	60-130		
AM	260-280			243-271	40-105	224-252	40-105	239-270	50-110	224-252	50-110	242-272	60-140	242-272	60-121	239-268	60-135	239-268	60-135
AN	280-300					243-271	40-105	260-289	50-110	239-270	50-110	262-290	60-140	262-290	60-121	258-287	60-135	258-287	60-135
AO	300-320									260-289	50-110	280-310	60-140	280-310	60-121	277-305	60-135	277-305	60-135
AP	320-340											300-328	60-140	300-328	60-121	295-324	60-135	295-324	60-135
AQ	340-360													318-347	60-121	314-342	60-135	314-342	60-135
AR	350-370																	323-352	60-135
AS	360-380																		

OPERATING CHARACTERISTICS AT 50Hz

H.P.U. = Approximate Pick-up at 90°C P.U. and D.O. values at 25°C

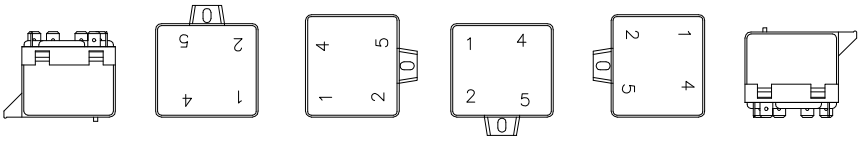
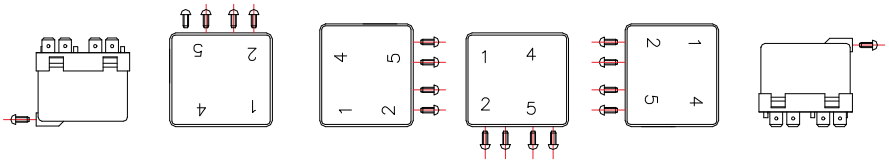
Coil number		549		550		554		555		558		556		559		560		561	
Vmax at 40°C (VAC)		151		299		300		338		340		356		379		400		450	
Resistance at 25°C (Ω) ±10%		1500		5600		5700		7500		7600		10000		10700		11000		13800	
	H.P.U.	P.U.	D.O.	P.U.	D.O.	P.U.	D.O.	P.U.	D.O.	P.U.	D.O.	P.U.	D.O.	P.U.	D.O.	P.U.	D.O.	P.U.	D.O.
A	120-130	111-124	20-45																
B	130-140	120-134	20-45	132-148	40-90														
C	150-160	130-144	20-45	140-153	40-90														
D	160-170	140-153	20-45	150-163	40-90														
E	170-180	149-163	20-45	162-175	40-90														
F	180-190			171-184	40-90	162-175	40-90					180-195	40-105						
G	190-200			180-193	40-90	171-184	40-90	180-195	40-105			189-205	40-105	180-195	60-121				
H	200-220			186-215	40-90	180-193	40-90	190-215	40-105	180-195	40-105	186-214	60-130	189-224	60-121				
I	220-240			205-234	40-90	186-215	40-90	208-239	50-110	190-215	40-105	204-233	60-130	204-233	60-121				
L	240-260			224-252	40-105	205-234	40-90	224-252	50-110	208-239	50-110	223-252	60-130	223-259	60-121	223-252	60-130		
M	260-280			243-271	40-105	224-252	40-105	239-270	50-110	224-252	50-110	242-272	60-140	242-272	60-121	239-268	60-135	239-268	60-135
N	280-300					243-271	40-105	260-289	50-110	239-270	50-110	262-290	60-140	262-290	60-121	258-287	60-135	258-287	60-135
O	300-320									260-289	50-110	280-310	60-140	280-310	60-121	277-305	60-135	277-305	60-135
P	320-340											300-328	60-140	300-328	60-121	295-324	60-135	295-324	60-135
Q	340-360													318-347	60-121	314-342	60-135	314-342	60-135
R	350-370																	323-352	60-135
S	360-380																		

If a 60Hz model is used at 50Hz, its 50Hz PU will be about 13% lower.
If a 50Hz model is used at 60Hz, its 60Hz PU will be about 13% higher

Product Characteristics

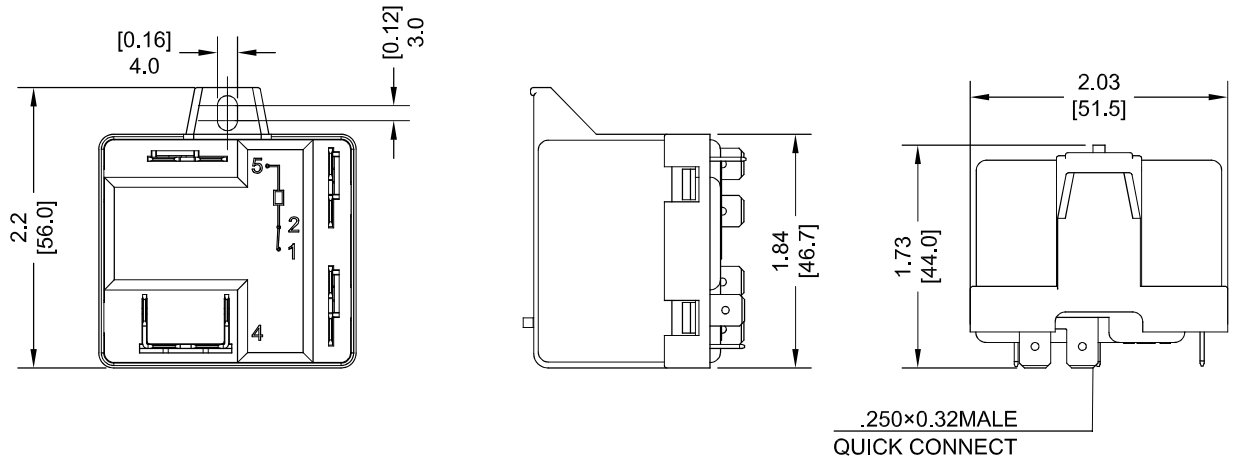
Maximum operating voltage		500VAC @-40℃~50℃
		500VAC @-40℃~85℃
Contact form		SPST-NC
Contact resistance		≤50mΩ(@24V/1A)
Rated load		16A 400VAC
Maximum commutation voltage		500VAC
Maximum commutation current		50A
Input power	Holding power	≤5 VA
	Input power	≤21.5 VA
Mechanical durability		$1X10^6$
Electrical durability		SPST-NC: 5 x 10^5OPS 16A on and off 400VAC cosØ=0.7-0.8 at 25 1s on 9s off SPST-NC: 2 x 10^5OPS 35A only off 400VAC cosØ=0.7-0.8 at 25 1s on 9s off SPST-NC: 1 x 10^5OPS 50A only off 400VAC cosØ=0.7-0.8 at 25 1s on 9s off
Insulation resistance		500MΩ（500VDC）
Shock（stability）		$98m/s^2$
Vibration		10Hz to 55Hz, 1.5mm double amplitude
Humidity range		5% to 85%RH
Temperature range		-40℃ to 85℃
Lead out form		Quick connect
Weight		About 98g
Packaging method		Dust cover type

Operating position

	1	2	3	4	5	6
TAB MOUNT						
PANEL MOUNT						

Dimensions

inch[mm]



Wiring diagram

